



Exploring different indicators for quantifying surface urban heat and cool island together: A case study over two metropolitan cities of India

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Abstract

The rise of urbanization has led to many critical issues like increased air pollution, emission of more greenhouse gases, sudden climate change, and a rise in temperature of an urban area compared to surrounding rural regions known as urban heat islands (UHI). It becomes crucial to understand the effect of climatic conditions in those areas. In this present research, we have chosen two different cities: Bhopal and Guwahati, situated over different climatic conditions. The study examines the diurnal, seasonal and inter annual variability of surface UHI (SUHI), using MODIS LST (MOD11A2 and MYD11A2) product of 1 km spatial resolution from 2003 to 2018. The temporal trend of LST is estimated using Mann–Kendall and Sen’s slope estimator test. This study also evaluates the land use land cover (LULC) change over the two cities using the LULC data. The present research also incorporates the determination of six different driving factors of SUHI starting from built-up area, urban–rural difference in evapotranspiration, black sky albedo, enhanced vegetation index, thermal inertia, and population. Pearson’s correlation criteria are used to determine the correlation of SUHI with each of the determinant parameters. The results reveal the presence of a cool urban island during the summer daytime in Bhopal, while Guwahati shows a positive SUHI. The nighttime shows clear evidence and higher SUHI magnitude than the daytime for Bhopal city. Contrary, a reverse phenomenon is evident in Guwahati with higher SUHI magnitude during the daytime compared to nighttime. The negative SUHI variation can be ascribed to rural area dynamics with the majority of crop-land cover. The evapotranspiration shows reasonable control of SUHI’s daytime variation. Thermal inertia was found to be an influential parameter in explaining the diurnal variation of SUHI.

Keywords Urban heat island · Urban cool island · Diurnal · Seasonal · MODIS · Land cover · Thermal inertia

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1 Introduction

Globally, more than one-half of the world's population (around 56.61%) lives in densely urbanized areas, which is expected to rise about 68% by the year 2050 (United Nations, 2018). Asian cities are the major hubs of their population and have shown tremendous growth in the last two decades, with around 52% of urbanites. Consequently, a profound challenge in the cities size and spatial structure to accommodate these overcome urban populations is seen over the coming decades. It is projected that around 66% of the population has to be accommodated by Asian cities by 2050 (United Nations, 2018). The cities of India currently houses nearly 34% of its total populace, and by the end of 2050, this number is projected to increase to more than 50% (United Nations, 2018). Cities are the world's major hub to accommodate the ever-growing population, so it is essential to assess how these areas will respond to rapid urbanization in the near future.

For millions of people living in dense metropolitan areas, increasing temperature due to rapid urbanization is a matter of concern. It can cause several environmental issues deteriorating the urban thermal environment. In recent times, one of such critical issues is the urban heat island (UHI) phenomenon, whereby an urban area experiences warmer temperatures than its rural surroundings (Oke, 1973; Voogt & Oke, 2003; Zhou et al., 2019). The prime cause of UHI is the transformation of pervious surfaces to the impervious surface, which alters the surface properties of the urban material (Kim & Brown, 2021a, 2021b; Rizwan et al., 2008). The effect of UHI can be seen from the cities of all sizes situated in different climatic regions (Heaviside et al., 2017). The season and climatic conditions of a city define the positive or negative consequences of UHI. It is seen that the cities with colder climates have positive feedback of UHI during the winter season as it reduces the heating costs and cold-related deaths. On the contrary, it negatively impacts human health and livelihood in the summertime, outweighing the wintertime benefits (Heaviside et al., 2017). Since most of the world's dense urbanized areas are situated over tropical climates, the negative impact of UHI is much larger as compared to positive effects. The effect of UHI is to increase heat-related problems like heatwave and heatstroke conditions in urban areas. Moreover, it also degrades air pollution and increases the energy demand needed for air-conditioning. It is also responsible for increasing greenhouse gas emissions, which contribute to global warming.

Satellite remote sensing provides thermal data, which is most frequently used to analyze and quantify the effect of UHI as compared to in situ and ground measurement (Falah et al., 2019; Mirzaei, 2015; Mukherjee et al., 2017; Veena et al., 2020; Weng, 2009; Zhou et al., 2019). The satellite-based study has many advantages over in situ-based research, as it can provide broader synoptic coverage and repeatable information about the area being investigated. Moreover, the UHI study at different spatial and temporal scales can be studied using satellite-based thermal data (Anniballe & Bonafoni, 2015). For example, MODIS sensors can provide land surface temperature (LST) data at 1 km resolution daily (both day and night), which makes it feasible to study the diurnal behavior of the surface UHI (SUHI) phenomenon (Anniballe et al., 2014; Clinton & Gong, 2013; Karimi et al., 2021; Khan et al., 2022; Mohammad & Goswami, 2020; Mohammad et al., 2022; Yao et al., 2020). The annual and seasonal SUHI variation analysis and associated determinants like temperature, precipitation, aerosol, vegetation, nighttime light, and population over 22 major Indian cities are assessed using MODIS land surface temperature data (Raj et al., 2020). Besides, many other researchers across the world using MODIS LST data to explore the SUHI phenomenon in different parts of the world, including Chinese cities (Li et al.,

2019a, 2019b; Peng et al., 2019; Yuanzheng et al., 2019; Zhou et al., 2014, 2015), South America cities (Wu et al., 2019), USA cities (Imhoff et al., 2010; Ramamurthy & Sangobanwo, 2016), Indian cities (Barat et al., 2021; Das et al., 2020; Kumar et al., 2017; Mohammad & Goswami, 2021b, 2021c; Shastri et al., 2017; Sultana & Satyanarayana, 2019) and many others.

Several studies focus on change in LULC dynamic on LST variability and SUHI phenomenon using high-resolution satellite data such as Landsat series of the satellite. The impact of LULC change and rapid urbanization enhancing SUHI over Lucknow city is studied using Landsat data from 2002 to 2014 (Singh et al., 2017). Their study suggests that cities exhibit higher LST than their rural surroundings, and a SUHI magnitude of 0.75 °C is found over the city. The SUHI phenomenon over six metropolises of India is assessed using the Landsat series of data at three different time intervals, each around five years from 2002 to 2013 (Sultana & Satyanarayana, 2019). The study of rural background conditions plays an essential role in determining the SUHI intensity, as revealed from a recent study of SUHI over Ahmedabad city, India, characteristic of negative SUHI during summer daytime, which is a common phenomenon for the cities of dry climatic regions (Bala et al., 2020; Mohammad & Goswami, 2021a; Mohammad et al., 2019). Other cities in India include Pune (Gohain et al., 2021), Guwahati (Borbora & Das, 2014), Mumbai (Rahaman et al., 2020), cities of Indo-Gangetic plains (Barat et al., 2021), Nagpur (Kotharkar & Bagade, 2018), Delhi (Pramanik & Punia, 2019; Yadav et al., 2017), Kolkata (Khan & Chatterjee, 2016), Chandigarh (Mathew et al., 2016), and many other evident the occurrence of SUHI.

Although many studies characterize the diurnal, seasonal, and temporal variation of SUHI over a particular single Indian city, comparative analysis of the urban cool island (UCI) and urban heat island (UHI) cities is required over two different climatic zones. It will provide information on the behaviors of cities with both UCI and UHI together, which will help to understand the mechanism and dynamics of UHI in both cases. In this study, we choose two cities of India, namely Bhopal and Guwahati, situated over different climatic conditions to study the diurnal and seasonal variation of SUHI during the last 16 years (2003 to 2018). This study also includes the detailed study of different determinates of SUHI like vegetation, albedo, evapotranspiration, and thermal inertia. Besides, the LULC analysis is also assessed, which affects the LST and thus the SUHI intensity. The comparative analysis of UCI and UHI will deliver the detail of SUHI and its different driving parameters, which would provide valuable information for urban policymakers and town planners to formulate the UHI mitigation strategies.

2 Study area

Bhopal, known as the city of lakes, is situated over central India in Madhya Pradesh (Fig. 1), at 23.25° N latitude and 77.41° E longitude. It is located toward the north of the Vindhya mountain ranges with undulating topography with an average elevation of about 427 m. The city is surrounded by two larger lakes known as the Bada Talab and Chota Talab. The cities geographic extend is about 463 sq. km, with a population of 2.4 million (United Nations, 2018). The city has around 138 sq. km of built-up land, while the surrounding rural region covers the majority with cropland (293.6 sq. km). It is categorized in a humid subtropical climate, with hot summer and mild dry winters. During the summer, the recorded mean temperature is around 30 °C, with the peak temperature in May, where

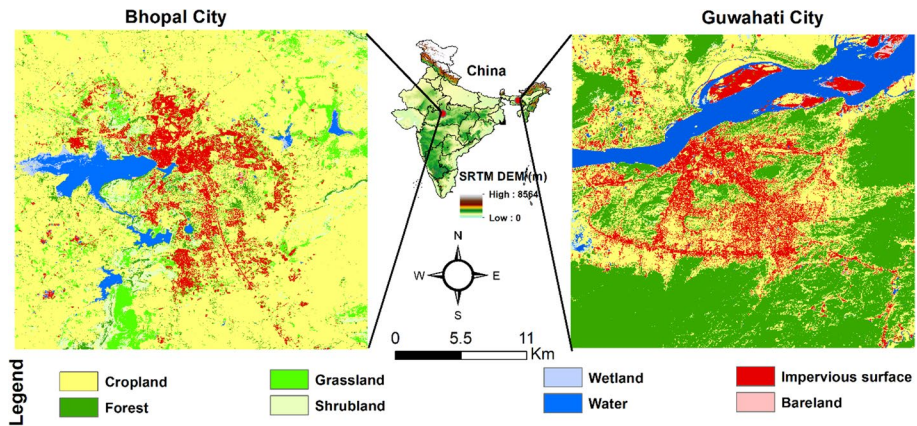


Fig. 1 Map of the study area showing the city of Bhopal and Guwahati (source, 10 m LULC from Gong et al., 2019)

the temperature exceeds 40 °C. In winter, it goes up to 18 °C, suggesting a mild, sunny, and dry season over the city. The city receives a total annual rainfall of about 1146 mm.

Guwahati, a gateway to northeast India, is a major riverine port city situated on the south bank of river Brahmaputra in Shillong plateau's foothills (Fig. 1). It is positioned at 26.10° N latitude and 92.49° E longitude with an altitudinal height of about 55.5 m above msl. The city has a total geographic area of approximately 127.84 sq. km with about 1.2 million population (United Nations, 2018). The city has around 119 sq. km of built-up land and 203.6 sq. km of cropland land in the surrounding region. The city is situated in a humid subtropical hot climate, where summers are hot and humid, while winters are mild and dry. The city's recorded mean temperature is about 24.2 °C, and the mean annual rainfall is around 1722 mm.

The two cities have been selected in this study because of their different rural area coverage (Fig. 1), which affects the SUHI intensity. The presence of cropland areas in the surrounding region of Bhopal city and greenery in Guwahati will play a key role in understating the diurnal variability of SUHI. The presence of water bodies near both cities also plays a vital role in modifying the SUHI phenomenon.

3 Data used

The analysis of SUHI and their driving parameters requires a large number of datasets from multiple sensors/sources over several years. First is the requirement of LST datasets, which is essential for estimating SUHI. In this study, the LST product of the MODIS sensor, onboard Terra and Aqua satellites is used. It measures the radiation from the earth's surfaces in 36 broad spectral bands between 0.4 and 14.4 μm. The Terra and Aqua satellite passes around 10:30 and 13:30 local solar time over the study location during daytime. Whereas in nighttime, it passes around 22:30 and 01:30 local solar time. The present study considered a global 8-day LST composite (MYD11A2 and MOD11A2) of 1 km spatial resolution from 2003 to 2018, stored in a sinusoidal grid format. The split-window algorithm is used to estimate LST from MODIS thermal bands under clear-sky conditions. The LST product is validated with the help of in situ temperature measurements, which

provides an error of less than 0.5 K in most cases (Wan, 1996, 2014). Only LST pixels of high quality were selected in this study, based on quality control (QC) of datasets, eliminating the effects from the retrieval algorithms errors and clouds. Each year, the daytime LST was generated by taking the mean of the 10:30 and 13:30 images, and nighttime LST was by considering the mean of 22:30 and 01:30 images, respectively. The study includes two different seasons separately. Summer season data were obtained from April, May, and June, whereas winter from December, January, and February.

In addition to LST, this study also considered data from different sensors and sources to analyze the SUHI determinants. For the period 2003–2018 data, six other variables were considered in this study: LULC, enhanced vegetation index (EVI), while sky albedo (WSA), evapotranspiration (ET), thermal inertia (TI), and population (Pop).

The annual global LULC data of the Climate Change Initiative of Europe Space Agency (CCI-ESA) with 300 m spatial resolution was used to extract the different land cover classes maps. EVI and WSA data were obtained from MOD13A3 (monthly composite) and MCD43C3 (16-day composite), respectively. EVI data were at 1 km spatial resolution and WSA at 500 m, which were resampled to 1 km. ET data were obtained from the SSEB (Simplified Surface Energy Balance) model, with a spatial resolution of 1 km (Senay et al., 2011). The model shows an inimitable parameterization required for operational application, i.e., each pixel has pre-defined distinguish boundary conditions. It was based on the original formulation of the hot/cold pixels principles of SEBAL (Bastiaanssen et al., 1998) and METRIC (Gowda et al., 2008) models. Moreover, TI data were also prepared from the day-night LST and WSA datasets at a spatial resolution of 1 km following the methodology of Mohammad & Goswami, 2021b. Furthermore, the population data obtained from World Urbanization Prospects (WUP) was also considered in this study to find the city's populous information.

4 Methodology

The methodology of the present study includes the spatial variability of LST from MODIS, its temporal trend analysis using Mann–Kendall and Sen's slope estimator test. The urban and rural boundary extracted from ESA's LULC maps helped determine the SUHI intensity. The variability of ESA based LULC maps were also discussed in this study, as the impact of land cover is essential in determining the SUHI. The variability of different considered drivers were also analyzed, and the interrelation of these variables with SUHI intensity is presented with the help of Pearson's correlation analysis. The overall methodology adopted in this present study is shown in Fig. 2.

4.1 Trend analysis method

Mann–Kendall (MK) test is used in this study to detect the trend in LST datasets. It is a non-parametric rank test initially developed by Mann and Kendall (Kendall, 1975; Mann, 1945). One of the major advantages of these tests is that they can quickly detect a trend in both linear and nonlinear datasets. The MK trend has been used extensively over a variety of time series and has proven to be an efficient tool to detect the trend (Malik et al., 2020; Mohammad & Goswami, 2019).

The MK test statistic S is calculated in time series data using Eq. 1 as follows:

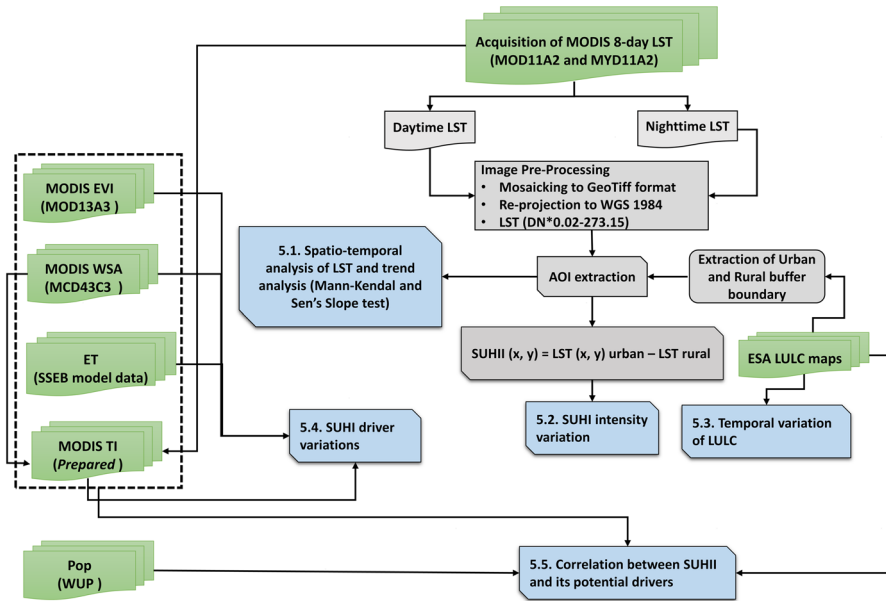


Fig. 2 Flowchart of the overall methodology adopted in this study

$$S = \sum_{i=1}^{N-1} \sum_{j=i+1}^N \text{sgn}(x_j - x_i) \quad (1)$$

where x_i and x_j are the data values of the input time series in the year i and j , N represents the total number of years in the time series. The sgn function in Eq. 1 is defined using Eq. 2 as follows

$$\text{sgn}(x_j - x_i) = \begin{cases} 1 & \text{if } (x_j - x_i) > 0 \\ 0 & \text{if } (x_j - x_i) = 0 \\ -1 & \text{if } (x_j - x_i) < 0 \end{cases} \quad (2)$$

The value of sgn function varies from 1 to -1, depending upon the value of $(x_j - x_i)$ as seen from the above equation. The variance of S ($\text{Var}(S)$) is calculated using Eq. 3 as shown as follows:

$$\text{Var}(S) = \frac{1}{18} \left[N(N-1)(2N+5) - \sum_{p=1}^q t_p(t_p-1)(2t_{p+5}) \right] \quad (3)$$

where t_p is the number of tie points in the time series for p th value, and q is the number of tied values present on the time series data. Based on the S from Eq. 1 and $\text{Var}(S)$ from Eq. 3, the final MK test statistics Z_{MK} is calculated as shown in Eq. 4

$$Z_{MK} = \begin{cases} \frac{S-1}{\sqrt{\text{Var}(S)}} & \text{if } S > 0 \\ 0 & \text{if } S = 0 \\ \frac{S+1}{\sqrt{\text{Var}(S)}} & \text{if } S < 0 \end{cases} \quad (4)$$

The value of Mann Kendall Z_{MK} tells us the present trend in the times series. The increasing time series is represented by positive Z_{MK} values, whereas decreasing time series is by negative Z_{MK} values. When the condition $|Z_{MK}| > Z_{1-\alpha/2}$, is satisfied, we can say that the null hypothesis is rejected in the time series, and a significant trend is present. $Z_{1-\alpha/2}$ is used to define the critical value of Z for a specific significant level. In this study, we considered the 5% significant level, which gives the value of $Z_{1-\alpha/2}$ to 1.96.

The magnitude of the trend is estimated using Sen's slope estimator test, a nonparametric method used to calculate the magnitude (slope) of a trend (Sen, 1968). It is calculated using Eq. 5:

$$Q_i = \frac{x_j - x_k}{j - k} \quad \text{for } i = 1, \dots, N \quad (5)$$

where x_j and x_k represent the data values for year j and k ($j > k$), respectively. N is the total number of data values in the time series. If there is only one datum in the time series in each time period having n as the total number of time periods, then $N = \frac{n(n-1)}{2}$. When the datum is multiple with more than one time period, then $< \frac{n(n-1)}{2}$.

The magnitude (slope) in the time series of N values is calculated using Eq. 6, considering the Q_i values to be sorted from smallest to largest.

$$Q_{med} = \begin{cases} Q_{[(N+1)/2]}, & \text{if } N \text{ is odd} \\ \frac{Q_{[N/2]} + Q_{[(N+2)/2]}}{2}, & \text{if } N \text{ is even} \end{cases} \quad (6)$$

where, the sign of Q_{med} reflects the trend reflection of the time series data, while its value indicates the steepness of the trend.

4.2 Estimation of surface urban heat island (SUHI) intensity

SUHI intensity (SUHII) is estimated as the difference between the mean LST of urban and their surrounding rural areas. This study calculated the rural reference LST as the mean LST of about 5 km buffers rural pixels from the outer urban periphery. The positive and negative values of SUHII represent the warming and cooling effect of the urban area, respectively. SUHII describes the thermal behavior of an urban area and is calculated using Eq. 7 as shown as follows:

$$SUHII(x, y) = LST(x, y)_{urban} - LST_{rural} \quad (7)$$

where (x, y) indicates the location (longitude, latitude) of a pixel. $LST(x, y)_{urban}$ represents the mean LST of urban pixels and LST_{rural} , as the mean LST of rural pixels within the buffer area.

4.3 Potential drivers of SUHI

The LULC changes significantly affect the SUHI effect over an urbanized area. The present study discusses the changes in LULC from 2000 to 2018 as a SUHI effecting parameter. The built-up class was considered as potential SUHI drivers in this study. Moreover, SUHI has also been substantially affected by vegetation, evapotranspiration, albedo, and thermal inertia. In this study, the difference between the urban and rural areas of EVI, ET, WSA, and TI (ΔEVI , ΔET , ΔWSA , and ΔTI) are considered as potential drivers affecting SUHI. The effect of the rising population was also included in the investigations as potential SUHI drivers. Pearson's correlation coefficient between the SUHI and the potential drivers was calculated to investigate the effect of each driver in SUHI over the years.

5 Results and Discussion

5.1 Spatial distribution of LST

The spatial distribution variation of mean LST over the Bhopal city in both summer/winter and daytime/nighttime is shown in Fig. 3. Additionally, this figure also indicates Sen's slope trend over the last 16 years. The result of LST suggests that the mean LST reaches up to 44.12 °C in the urban area and 44.64 °C in the rural area during summer daytime, whereas during winter daytime it shows a value of 29.70 °C and 28.76 °C in the urban and rural area, respectively (Fig. 4). On the other hand, nighttime witness lower LST than daytime. The recorded mean LST during summer nighttime in urban and rural area is 25.36 °C and 24.14 °C respectively, while in winter nighttime it goes up to 14.13 °C in the urban and 12.65 °C in the rural area. The mean LST of urban areas exceeds the rural ones during nighttime (both summer and winter season), suggesting

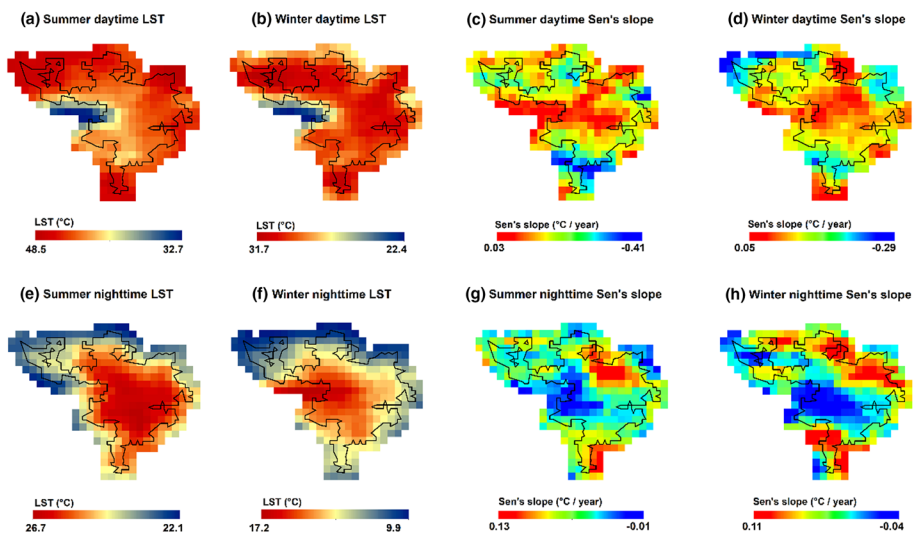


Fig. 3 Spatial distribution and the mean LST trend during 2003–2018 over Bhopal city

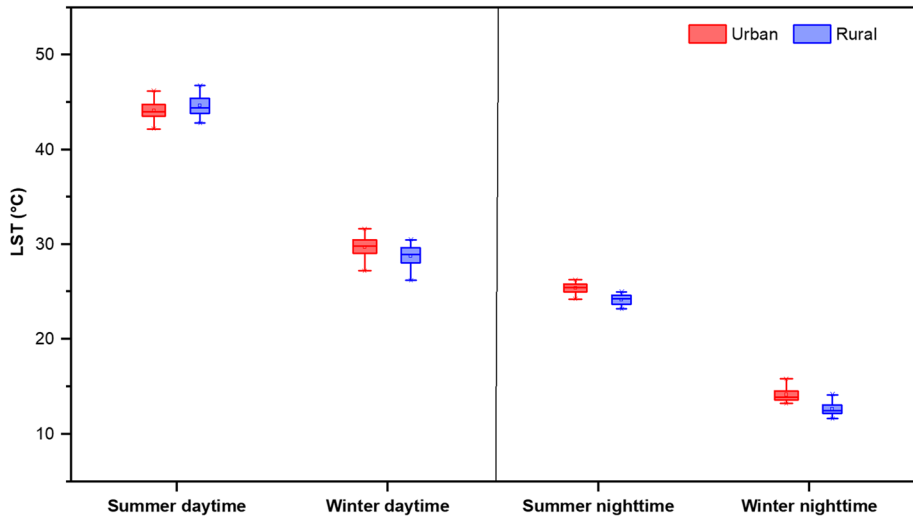


Fig. 4 Variation of LST (daytime and nighttime) of Bhopal city's urban and surrounding rural areas during 2003–2018

a positive SUHI. On the contract during the daytime, both positive/negative SUHI is expected, positive in winter daytime and negative in summer daytime (Fig. 4).

The spatial patterns of LST during daytime (Fig. 3a, b) reveal a higher LST in the surrounding rural area, suggesting a negative SUHI. Whereas in the nighttime, a significant positive SUHI is evident with the urban area showing higher LST than the surrounding rural area (Fig. 3e, f).

The trend and slope of the LST data during the last 16 years are estimated from Mann–Kendall and Sen's slope estimator tests (Fig. 3c, d, g, h). The result of Sen's slope during daytime suggests that the trend value ranges between 0.03 and -0.41 °C/y for summer and 0.05 and -0.29 °C/y for winter. Whereas in nighttime, the trend vary between 0.13 and -0.01 °C/y for summer and 0.11 and -0.04 °C/y for winter.

The spatial distribution (diurnal and seasonal) of mean LST over the last 16 years for Guwahati city, is shown in Fig. 5. The result suggests that the mean LST reaches up to 33.24 °C in the urban area and 30.98 °C in the rural area during summer daytime, while in winter daytime it shows a value of 25.93 °C and 24.75 °C for urban and rural areas, respectively, as shown in Fig. 6. On the other hand, mean LST during summer nighttime is 22.32 °C for urban and 21.68 °C for the rural area, while it is 15.32 °C and 14.67 °C for urban and rural areas in winter nighttime (Fig. 6). Overall, the mean LST in urban areas exceeds the rural area during daytime/nighttime and summer/winter, prefiguring a positive SUHI effect over the Guwahati city.

The mean LST from the spatial pattern distribution during the daytime LST (Fig. 5a, b) and nighttime (Fig. 5e, f) exhibits a higher LST of the urban area with respect to the surrounding rural area. This suggests positive SUHI during both summer /winter and daytime/ nighttime over Guwahati city.

The Mann–Kendall and Sen's slope estimator tests are used to determine the trend and slope of LST data from 2003 to 2018 over the Guwahati city (Fig. 5c, d, g, h). The result of Sen's slope during the daytime reveals that the trend ranges between 0.29 and -0.04 °C/y

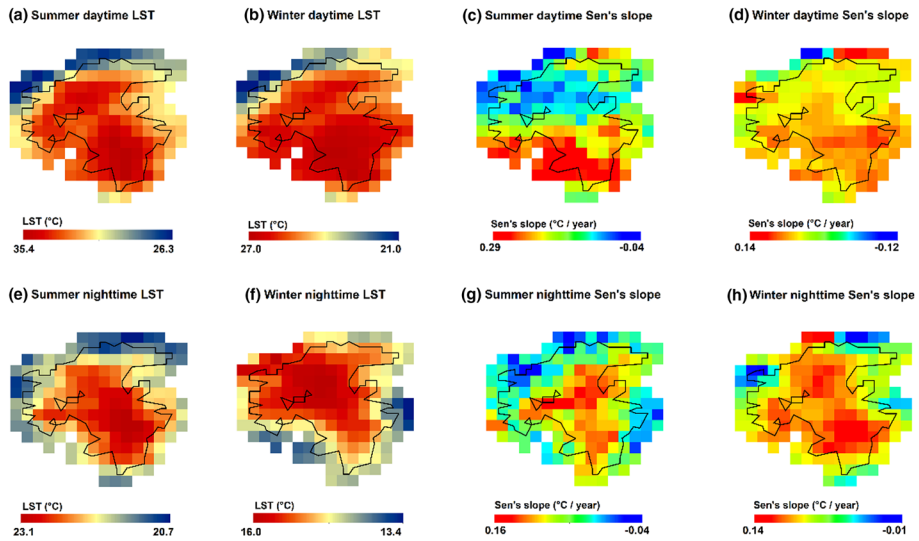


Fig. 5 Spatial distribution and the trend of the mean LST during 2003–2018 over Guwahati city

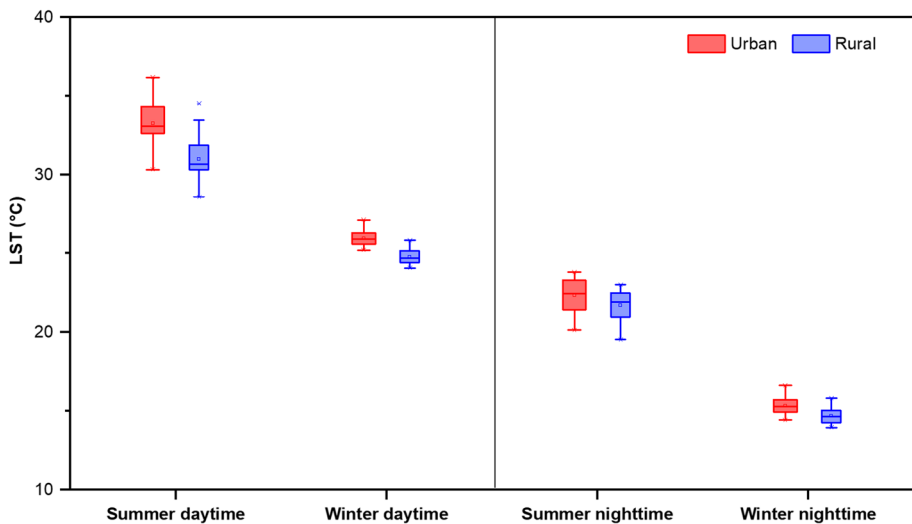


Fig. 6 Variation of LST (daytime and nighttime) of Guwahati city's urban and surrounding rural area during 2003–2018

for summer and 0.14 and -0.12 $^{\circ}\text{C}/\text{y}$ for winter, whereas in the nighttime, the ranges vary between 0.16 and -0.04 $^{\circ}\text{C}/\text{y}$ for summer and 0.14 and -0.01 $^{\circ}\text{C}/\text{y}$ for winter.

The result of diurnal and seasonal LST variation reveal that the mean urban LST is higher than the rural LST in all the seasons, except in summer daytime, where the behavior is opposite. Bhopal witnessed lower LST over urban areas, i.e., urban cool island phenomenon, whereas Guwahati shows higher urban LST, i.e., urban heat island phenomenon. The UHI effect in areas with vegetation cover and lighter surface color of buildings

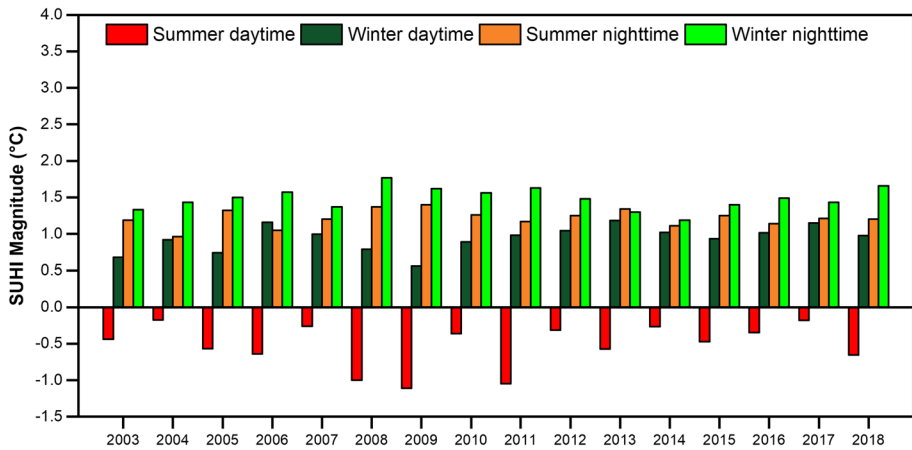


Fig. 7 Temporal variation of SUHI magnitude from 2003 to 2018 during daytime/nighttime and summer/winter over Bhopal city

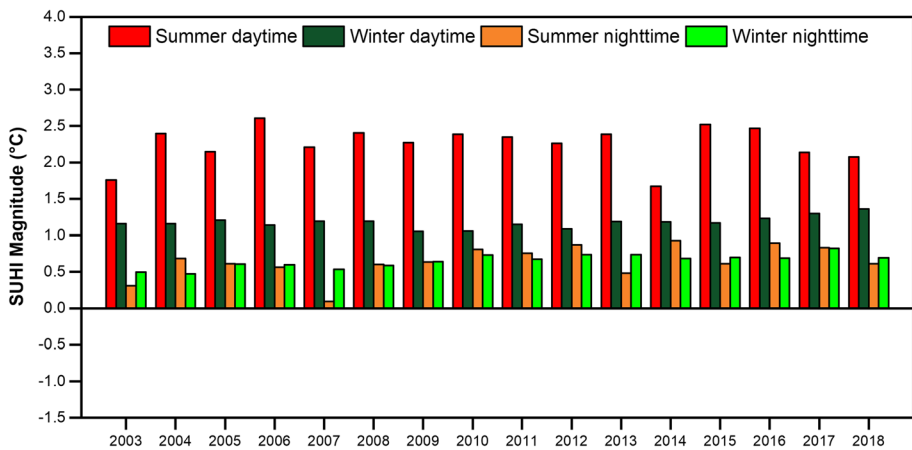


Fig. 8 Temporal variation of SUHI magnitude from 2003 to 2018 during daytime/nighttime and summer/winter over Guwahati city

is significantly less as compared to the surrounding rural areas (Gupta et al., 2009). The negative UHI in the daytime is also reported in recent studies, like cities along the Gangetic Plain of India (Barat et al., 2021), cities of dry humid climatic conditions of India (Mohammad & Goswami, 2021b), and many others.

5.2 SUHI magnitude

The variation in mean SUHI magnitude (both diurnally and seasonally), calculated from the difference in mean LST of the urban and rural area, is shown in Fig. 7 for Bhopal city and Fig. 8 for Guwahati city over the study period.

The result reveals that the Bhopal city witnessed a consistently negative mean SUHI magnitude of -0.52°C during summer daytime over the last 16 years (Fig. 7), whereas in winter daytime, the mean SUHI magnitude was 0.94°C . During the nighttime, the mean SUHI magnitude was 1.21°C in summer and 1.48°C in winter (Fig. 7). Overall, the SUHI was more prominent during nighttime than daytime in both the seasons. Similar behavior of SUHI variation is also seen over the Ahmedabad city attributed to dry climatic conditions (Mohammad et al., 2019).

During summer daytime, a prominent higher positive mean SUHI is seen over the Guwahati city (Fig. 8). The result suggests that the mean SUHI magnitude during the summer daytime was 2.25°C over the last 16 years, while winter daytime shows a comparatively lower SUHI magnitude of 1.18°C , as compared to summer daytime. Compared to daytime, the SUHI effect is less prominent in the nighttime during both seasons, with a mean magnitude of 0.64°C in summer and 0.65°C in winter over the city. The surrounding of the Guwahati city is mainly covered with a green cover, which provides natural cooling and lowers the LST resulting in higher SUHI during daytime (Mohammad & Goswami, 2021c). Borbora and Das (2014) also reported the occurrence of negative SUHI in summer daytime over Guwahati city.

5.3 Temporal variation of land use land cover

The LULC variation of Bhopal city during the selected study period is shown in Fig. 9. The city witnessed a rising trend of built-up area (or impervious area) from 54.9 km^2 in 2000 to 138.0 km^2 in 2018. The surrounding rural area covers the majority of cropland, wetland, and forest, showing a decreasing trend over the study period. Cropland area showed a falling trend over the years from 305.4 km^2 in 2000 to 293.6 km^2 in 2018; wetland from 176.1 km^2 in 2000 to 166.7 km^2 in 2018 and forest from 123.6 km^2 in 2000 to 122.7 km^2 in 2018 (Fig. 9b). Grassland, also present in the nearby vicinity of the study area, shows a lower decreasing value from 96.8 km^2 in 2000 to 92.4 km^2 in 2018. This is a clear indication of alternation of other LULC classes to built-up classes due to the rapid urban growth. The city of Bhopal has a significant number of water bodies present, which show an increasing trend from 59.5 km^2 in 2000 to 63.4 km^2 in 2018. This may be the probable reason for having lower SUHI present over the city. Ghosh, 2019, also witnesses a rapid rate of horizontal expansion of the city area from 15.8 km^2 in 1991 to 184.5 km^2 in 2016 at an average change of $7\text{ km}^2\text{ year}^{-1}$. A recent study by Sharma et al., 2021, suggests that the built-up area showed the highest increase by 8576 ha (almost tenfolds from 1988 to 2018). This drastic change is responsible for the changing structure of the Bhopal city ecosystems.

Figure 10 shows the LULC variation of Guwahati city over the study period. The city clearly witnessed a rising trend of built-up area (or impervious area) from 27.7 km^2 in 2000 to 119.0 km^2 in 2018. On the other hand, the surrounding rural cover, consisting of cropland, wetland, and forest, shows decreasing trends over the study period. Cropland showed a falling trend over the years from 220.2 km^2 in 2000 to 203.6 km^2 in 2018; wetland from 84.2 km^2 in 2000 to 76.8 km^2 in 2018 and forest from 69.1 km^2 in 2000 to 63.2 km^2 in 2018 (Fig. 10b). Grassland, also present in the nearby vicinity, shows a lower decreasing value from 24.7 km^2 in 2000 to 25.3 km^2 in 2018. This is due to the rapid urbanization, which leads to the transformation of these vegetation and cropland areas to dense built-up area. A recent study also shows the decrease in these land covers over the Guwahati city at the expense of increasing built-up area (Nath et al., 2021). Bareland is also present in small

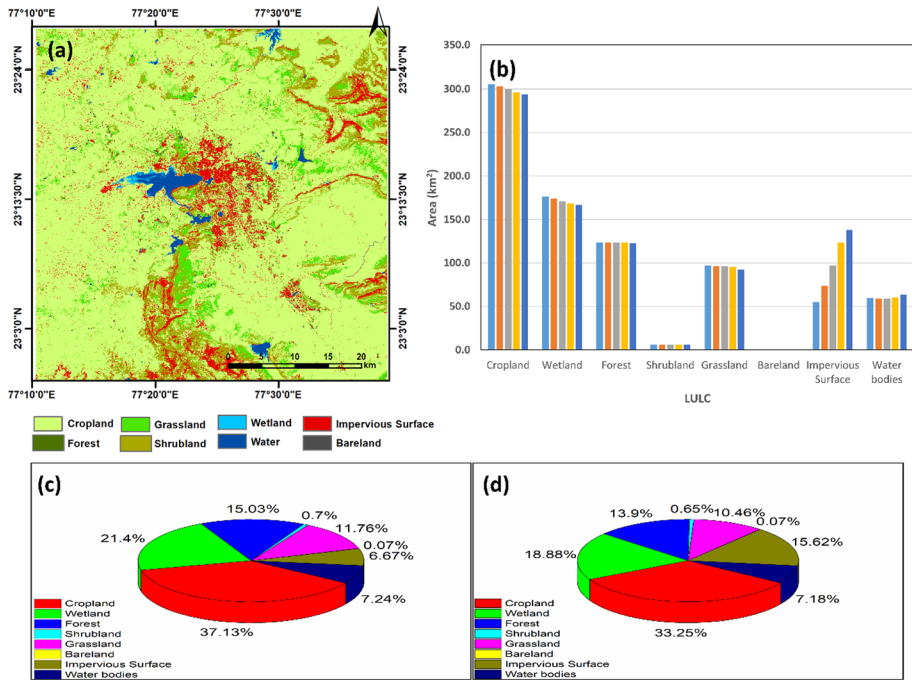


Fig. 9 Interannual variation (2000–2018) of the LULC of Bhopal city **a** spatial variation for the year 2018, **b** graphical variation for the years 2000, 2004, 2008, 2012, and 2018, **c** pie chat distribution for the year 2000 and **d** pie chat distribution for the year 2018

amounts showing an increasing trend from 27.7 km² in 2000 to 47.3 km² in 2018. Guwahati has a significant larger river known as the Brahmaputra river, which shows a remarkable decreasing trend from 108.8 km² in 2000 to 85.9 km² in 2018. This may be the probable reason for significantly higher SUHI present over the city in the daytime compared to nighttime during both the summer and winter seasons.

5.4 SUHI driver's variation

The relationships between the magnitude of SUHI and its potential driving factors were also examined in this study. This will help to understand each driver's contribution to the SUHI phenomenon over the cities of two different climatic zones. In this study, we have chosen built-up area, the urban–rural difference between enhanced vegetation index, evapotranspiration, black sky albedo, and thermal inertia (Δ EVI, Δ ET, Δ BSA, and Δ TI), and the population as the driving variable for SUHI. The distribution of built-up areas for the two chosen cities was discussed in the previous section. In this section, we will discuss the spatial and seasonal variation of Δ EVI, Δ ET, Δ BSA, Δ TI of these cities.

Figure 11 shows the spatial variation of mean EVI, ET, BSA, and TI for Bhopal city from 2003 to 2018 in summer/winter season. The urban area has a significantly lower EVI value than the rural area. The WSA of urban built-up areas is comparatively lower. Instead, a considerably higher albedo is observed in the SW part of the rural area due to a large lake's presence over the area. ET also shows higher value along with SW parts which

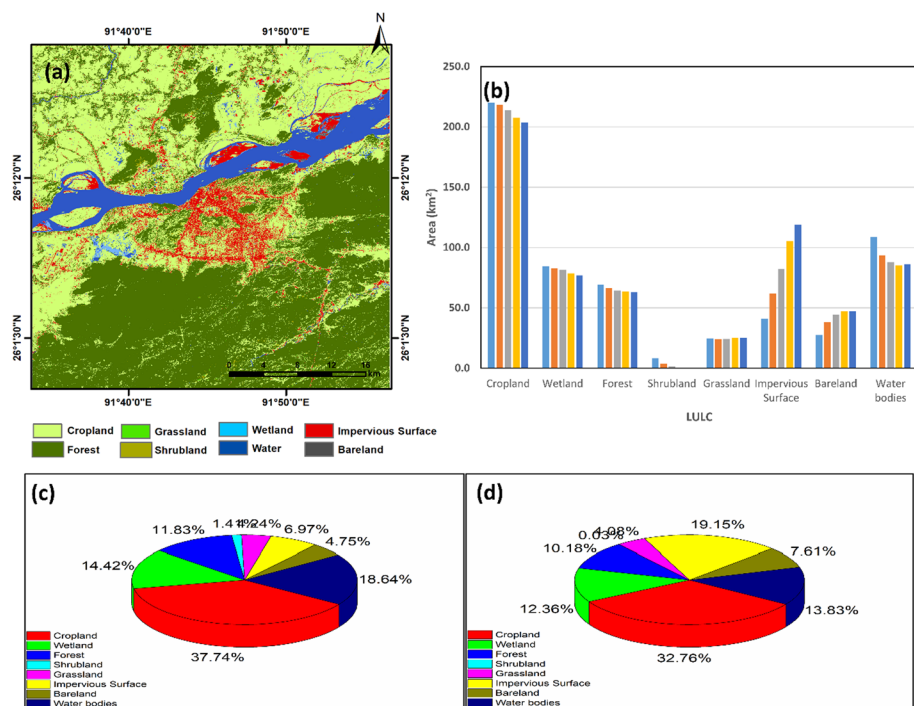


Fig. 10 Interannual variation (2000–2018) of the LULC of Guwahati city **a** spatial variation for the year 2018, **b** graphical variation for the years 2000, 2004, 2008, 2012, and 2018, **c** pie chat distribution for the year 2000 and **d** pie chat distribution for the year 2018

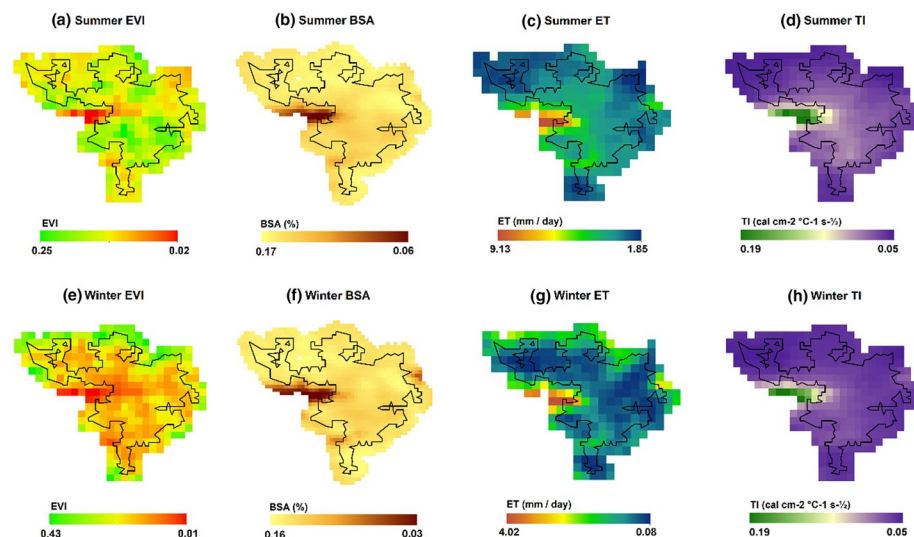


Fig. 11 Spatial variation of mean EVI, BSA, ET, and TI of Bhopal city during summer and winter seasons

substantially comes from the presence of lake over there. The thermal inertia shows a similar behavior prevailing lower value over urban and lake areas.

The spatial distribution of mean EVI, ET, BSA, and TI during the summer and winter seasons for Guwahati city is shown in Fig. 12. The lower EVI value is seen in the urban area than the suburban area. A significantly lower EVI value is seen over the NW rural area because of the Brahmaputra river. The albedo of the urban area is comparatively lower than rural areas, and a significantly higher albedo over the river is seen. The ET shows similar behavior as that of EVI and BSA, witnessing higher ET along the river channel.. The TI of urban land cover shows higher value than the rural land covers.

5.5 Correlation between SUHI and its potential drivers

This section includes the analysis of the causes of SUHI variability, considering six different parameters: urban built-up area), Δ EVI, Δ BSA, Δ ET, Δ TI (i.e., the urban–rural difference in EVI, BSA, ET, TI), and population (Pop). The variability of SUHI with this selected parameter is discussed in this section during daytime and nighttime separately in both summer/winter seasons.

Pearson's correlation between SUHI and its driving parameters is shown in Fig. 13 for Bhopal city. A weak correlation is seen during summer daytime except for two parameters, Δ ET and Δ TI, which showed a significantly higher correlation of -0.67 and -0.87 , respectively. Conversely, summer nighttime shows a negative correlation with Δ BSA (-0.22) and a strong positive correlation with Δ TI (0.42), and a small negative with Δ ET (0.21).

In winter daytime, SUHI exhibits a significantly higher correlation behavior in comparison to summer nighttime (except Δ TI), with a high value for Δ ET (-0.86) and Δ TI (-0.66). Conversely, winter nighttime SUHI has significantly higher correlations to summer nighttime with a profoundly good correlation in Δ TI ($r=0.84$) and Δ EVI ($r=-0.68$).

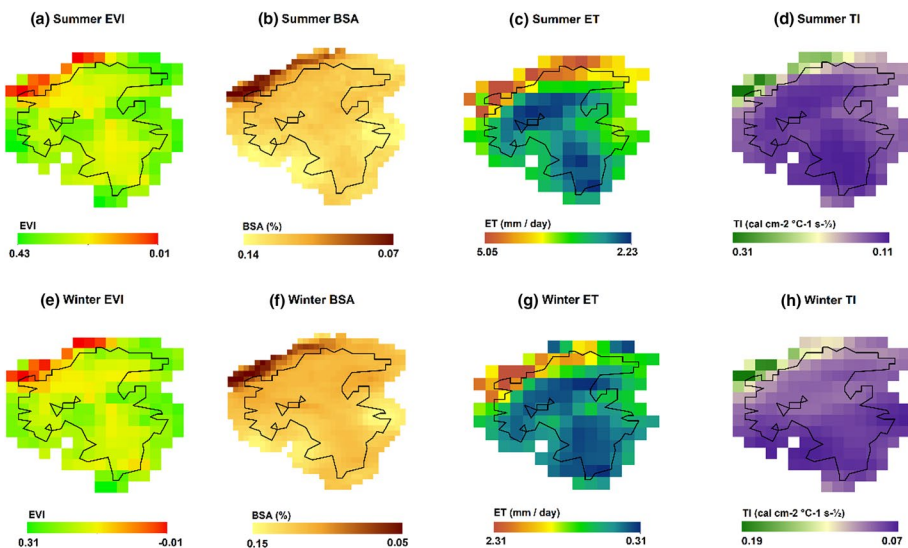


Fig. 12 Spatial variation of mean EVI, BSA, ET, and TI of Guwahati city during summer and winter

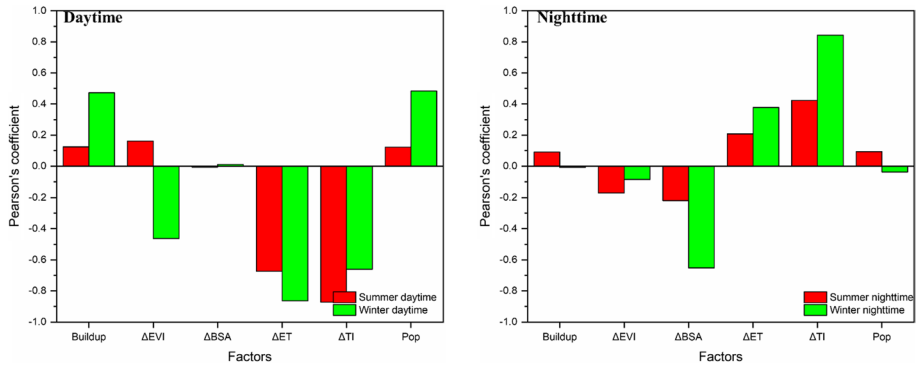


Fig. 13 Pearson's correlation coefficient relation between the magnitude of SUHI and its different causative potential drivers during both daytime and nighttime, in summer and winter season over Bhopal city

Despite having significant SUHI magnitude values, the correlations were weak in as compared to summer nighttime.

Pearson's correlation plot between SUHI and its potential driving parameters over Guwahati city during daytime/nighttime and summer/winter is shown in Fig. 14. During summer daytime, a very weak correlation is observed except ΔET ($r = -0.79$) and a very small correlation with ΔEVI (-0.31) and ΔTI (-0.22). On the contrary, summer nighttime SUHI is negatively correlated with ΔEVI and to some extent to ΔBSA and ΔET , whereas a significant positive correlation is seen with built-up (0.51), ΔTI (0.65), and Pop (0.50).

During winter daytime, a higher correlation is observed between the SUHI and the driving parameters with a positive correlation with built-up (0.48) and Pop (0.51), while others show a strong negative correlation with ΔET (-0.71) and ΔTI (-0.72) and a lower negative correlation with ΔBSA (-0.49) and ΔEVI (-0.38). Conversely, winter nighttime shows a strong positive correlation with built-up (0.84), Pop (0.82), and ΔTI (0.44), while ΔEVI and ΔBSA showed a significant negative correlation with SUHI with a value of -0.78 and -0.57 , respectively.

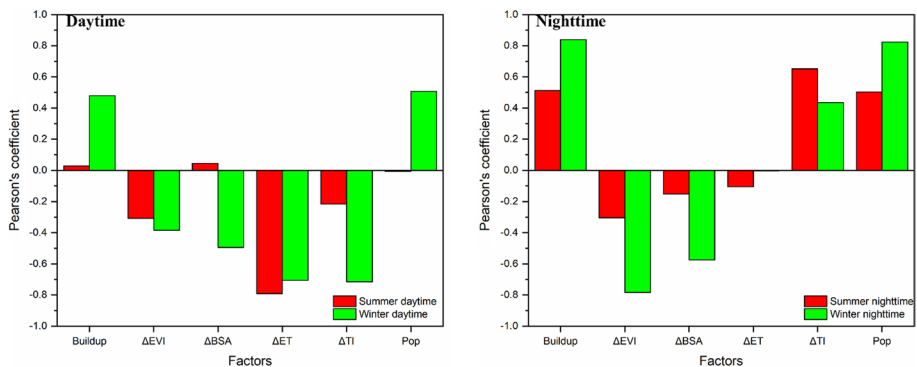


Fig. 14 Pearson's correlation coefficient relation between the magnitude of SUHI and its different causative potential drivers during both daytime and nighttime, in summer and winter season over Guwahati city

The urban area holds much heat compared to a rural area because urban surfaces trap more heat than the rural surface (Mehrotra et al., 2021; Oke, 1988). The built-up area shows a higher correlation than for the urban heat island city of Guwahati. More intense SUHI occurs in the nighttime as the release of heat from the stored urban ground surface is maximum in the nighttime (Li et al., 2011; Yuan & Bauer, 2007). The vegetation present in the rural area increases the latent heat via transpiration, resulting in lower LST as it cools the surface (Li et al., 2019a, 2019b; Peng et al., 2012). The findings of this study also confirm this phenomenon as ΔEVI shows a higher negative correlation with SUHI, especially in Guwahati city positive heat island. The ΔET shows a negative correlation with SUHI, which reveals that the water and vegetation provides cooling to the rural area through the process of evaporation and transpiration respectively (Karimi et al., 2021; Mohammad & Goswami, 2021b; Peng et al., 2019). The result of this study divulges that the effect of evapotranspiration is much more dominant in the daytime than nighttime. An increase in urbanization results in a decrease in the surface albedo and an increase in the heat storage of urban materials (Gachkar et al., 2021; Karimi et al., 2020; Kim & Brown, 2021a, 2021b; Mohammad et al., 2021; Stewart & Oke, 2012). Since most of the heat storage area is released during nighttime and contributes to nighttime SUHI, resulting in the higher negative correlation of SUHI with ΔWSA , as also seen from our present study. The capacity of a material to respond to heat gain and loss is represented by thermal inertial, and it represents the change in the ground heat flux of an area (Chen et al., 2008). The urban material has a higher thermal capacity than rural areas. So, ΔTI shows a positive correlation with SUHI during nighttime as stored heat is released during the night. In contrast, a reverse situation prevails during the daytime with a negative correlation (Mohammad & Goswami, 2021b). This is because the temperature of the urban surfaces can not rise rapidly as compared to rural surfaces.

6 Conclusion

The present research includes the investigation of spatial variability of LST and SUHI over the Bhopal and Guwahati city from 2003 to 2018, situated over two different climatic regions. The result of the spatial distribution of LST confirms the presence of a cool urban island over the Bhopal city and an positive UHI over the Guwahati city during summer daytime. Although the magnitude is small during the winter daytime, both cities reveal positive SUHI. The nighttime result suggests a positive heat island over the cities during both the seasons. The presence of cool urban islands in Bhopal city during summer daytime is due to cropland in a surrounding rural area and lower vegetation activity resulting in higher LST. The estimation of SUHI suggests a consequence of urban and rural area's warming/cooling dynamic. The evapotranspiration and vegetation are effective drivers in daytime SUHI variation in both seasons. The thermal inertia shows strong diurnal control on SUHI over both the cities in both summer/winter seasons. The thermal inertia analysis confirms it to be a more influential driver than other used drivers in this study. The result presented in this study in determining the seasonal and diurnal dynamics of LST and SUHI over two different Indian cities situated over different climatic conditions considering various determinant factors will help urban planners and policymakers to formulate mitigation strategies of the SUHI effect.

Though the study discusses the diurnal and seasonal variation of SUHI using MODIS land surface temperature data, other limitations and uncertainties remain in the study. The

limitation of MODIS overpass time over the study area does not include before and after sunrise and sunset to assess the highest SUHI at night and select the exact time of turn from daytime SUCI to nighttime SUHI. Therefore, this research has some weaknesses due to the limited satellite view of the ground and emissivity of different ground materials. It needs a more in-depth investigation of satellite characterization of urban areas to understand the mechanism and dynamics of SUHI better. Future research should consider higher spatial resolution data like Landsat to characterize the micro-level SUHI study over the city. This study was limited to assessing all possible satellite-based influencing drivers of SUHI. More factors that affect canopy UHI like precipitation, wind speed, urban geometry, soil types, and building components can be explored to understate SUHI better. The development of proper heat-related mitigation strategies is of utmost necessity for the better future sustainability of the selected urbanized area.

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Declarations

Declaration of Competing Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Consent for publication I give my consent for this paper to be published in your journal if found publishable.

Ethics approval and consent to participate I give my ethical approval for this paper to be published in your journal if found publishable.

References

- Anniballe, R., & Bonafoni, S. (2015). A stable gaussian fitting procedure for the parameterization of remote sensed thermal images. *Algorithms*. <https://doi.org/10.3390/a8020082>
- Anniballe, R., Bonafoni, S., & Pichierri, M. (2014). Spatial and temporal trends of the surface and air heat island over Milan using MODIS data. *Remote Sensing of Environment*, 150, 163–171. <https://doi.org/10.1016/j.rse.2014.05.005>
- Bala, R., Prasad, R., & Yadav, V. P. (2020). A comparative analysis of day and night land surface temperature in two semi-arid cities using satellite images sampled in different seasons. *Advances in Space Research*, 66(2), 412–425. <https://doi.org/10.1016/j.asr.2020.04.009>
- Barat, A., Partha Sarthi, P., Kumar, S., Kumar, P., & Sinha, A. K. (2021). Surface Urban Heat Island (SUHI) over riverside cities along the gangetic plain of India. *Pure and Applied Geophysics*. <https://doi.org/10.1007/s00024-021-02701-6>
- Bastiaanssen, W. G. M. G. M., Menenti, M., Feddes, R. A., & Holtslag, A. A. M. (1998). A remote sensing surface energy balance algorithm for land (SEBAL): 1. Formulation. *Journal of Hydrology*, 212–213(1–4), 198–212. [https://doi.org/10.1016/S0022-1694\(98\)00253-4](https://doi.org/10.1016/S0022-1694(98)00253-4)
- Borbor, J., & Das, A. K. (2014). Summertime Urban Heat Island Study for Guwahati City, India. *Sustainable Cities and Society*, 11, 61–66. <https://doi.org/10.1016/j.scs.2013.12.001>
- Chen, Y., Mingyi, Du., & Dong, R. (2008). Correlation between urban heat island effect and the thermal inertia using aster data in Beijing, China. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences XXXVII*, B7, 1723–1728.
- Clinton, N., & Gong, P. (2013). MODIS detected surface urban heat islands and sinks: global locations and controls. *Remote Sensing of Environment*, 134, 294–304. <https://doi.org/10.1016/j.rse.2013.03.008>

- Das, P., Sandeep Vamsi, K., & Zhenke, Z. (2020). decadal variation of the land surface temperatures (LST) and Urban Heat Island (UHI) over Kolkata City projected using MODIS and ERA-Interim dataSets. *Aerosol Science and Engineering*, 4(3), 200–209. <https://doi.org/10.1007/s41810-020-00067-1>
- Falah, N., Karimi, A., & Harandi, A. T. (2019). Urban growth modeling using cellular automata model and AHP (Case Study: Qazvin City). *Modeling Earth Systems and Environment*, 6(1), 235–248. <https://doi.org/10.1007/s40808-019-00674-z>
- Gachkar, D., Taghvaei, S. H., & Norouzian-Maleki, S. (2021). Outdoor thermal comfort enhancement using various vegetation species and materials (Case Study: Delgosha Garden, Iran). *Sustainable Cities and Society*, 75(August), 103309. <https://doi.org/10.1016/j.scs.2021.103309>
- Ghosh, S. (2019). A city growth and land-use/land-cover change: a case study of Bhopal, India. *Modeling Earth Systems and Environment*, 5(4), 1569–1578. <https://doi.org/10.1007/s40808-019-00605-y>
- Gohain, K. J., Mohammad, P., & Goswami, A. (2021). Assessing the impact of land use land cover changes on land surface temperature over Pune City, India. *Quaternary International*, 575–576(2020), 259–69. <https://doi.org/10.1016/j.quaint.2020.04.052>
- Gong, P., Liu, H., Zhang, M., Li, C., Wang, J., Huang, H., Clinton, N., Ji, L., Li, W., Bai, Y., Chen, B., Xu, B., Zhu, Z., Yuan, C., Suen, H. P., Guo, J., Xu, N., Li, W., Zhao, Y., et al. (2019). Stable classification with limited sample: Transferring a 30-m resolution sample set collected in 2015 to mapping 10-m resolution global land cover in 2017. *Science Bulletin*, 64(6), 370–73. <https://doi.org/10.1016/j.scib.2019.03.002>
- Gowda, P. H., Chávez, J., Howell, T. A., Marek, T. H., & New, L. L. (2008). Surface energy balance based evapotranspiration mapping in the Texas high plains. *Sensors*, 8(8), 5186–5201. <https://doi.org/10.3390/s8085186>
- Gupta, A., Vivek, D., & Aditya G. (2009). Mitigation of thermal pollution to enhance Urban air quality through remote sensing and GIS. In *10th ESRI India user conference*, (pp. 1–7).
- Heaviside, C., Macintyre, H., & Vardoulakis, S. (2017). The Urban heat island: Implications for health in a changing environment. *Current Environmental Health Reports*, 4(3), 296–305. <https://doi.org/10.1007/s40572-017-0150-3>
- Imhoff, M. L., Zhang, P., Wolfe, R. E., & Bounoua, L. (2010). Remote sensing of the Urban heat island effect across Biomes in the continental USA. *Remote Sensing of Environment*, 114(3), 504–513. <https://doi.org/10.1016/j.rse.2009.10.008>
- Karimi, A., Sanaieian, H., Farhadi, H., & Norouzian-Maleki, S. (2020). Evaluation of the thermal indices and thermal comfort improvement by different vegetation species and materials in a Medium-Sized Urban Park. *Energy Reports*, 6, 1670–1684.
- Karimi, A., Mohammad, P., Gachkar, S., Gachkar, D., García-Martínez, A., Moreno-Rangel, D., & Brown, R. D. (2021). Surface Urban heat island assessment of a cold desert city: A case study over the Isfahan metropolitan Area of Iran. *Atmosphere*, 12(10), 1368. <https://doi.org/10.3390/atmos12101368>
- Kendall, M. G. (1975). *Rank correlation methods*. Charles Griffin.
- Khan, F., Bhumika D., & Pir M. (2022). Urban growth modeling and prediction of land use land cover change over Nagpur City, India using cellular automata approach.” In *Geospatial technology for landscape and environmental management*, (pp. 261–82). Springer, Singapore. https://doi.org/10.1007/978-981-16-7373-3_13.
- Khan, A., & Chatterjee, S. (2016). Numerical simulation of Urban Heat Island Intensity under Urban – Suburban surface and reference site in Kolkata, India. *Modeling Earth Systems and Environment*, 2(2), 1–11. <https://doi.org/10.1007/s40808-016-0119-5>
- Kikon, N., Singh, P., Singh, S. K., & Vyas, A. (2016). Assessment of Urban Heat Islands (UHI) of Noida City, India using multi-temporal satellite data. *Sustainable Cities and Society*, 22, 19–28. <https://doi.org/10.1016/j.scs.2016.01.005>
- Kim, S. W., & Brown, R. D. (2021a). Urban Heat Island (UHI) Variations within a city boundary: A systematic literature review. *Renewable and Sustainable Energy Reviews*, 148, 111256.
- Kim, Y. J., & Brown, R. D. (2021b). A multilevel approach for assessing the effects of microclimatic urban design on pedestrian thermal comfort: The high line in New York. *Building and Environment*, 205(July), 108244. <https://doi.org/10.1016/j.buildenv.2021.108244>
- Kotharkar, R., & Bagade, A. (2018). Evaluating Urban heat island in the critical local climate zones of an Indian City. *Landscape and Urban Planning*, 169(August 2017), 92–104. <https://doi.org/10.1016/j.landurbplan.2017.08.009>
- Kumar, R., Mishra, V., Buzan, J., Kumar, R., Shindell, D., & Huber, M. (2017). Dominant control of agriculture and irrigation on Urban heat island in India. *Scientific Reports*, 7(1), 1–10. <https://doi.org/10.1038/s41598-017-14213-2>

- Li, J., Conghe Song, Lu., Cao, F. Z., Meng, X., & Jianguo, Wu. (2011). Impacts of landscape structure on surface Urban Heat Islands: A case study of Shanghai, China. *Remote Sensing of Environment*, 115(12), 3249–3263. <https://doi.org/10.1016/j.rse.2011.07.008>
- Li, D., Liao, W., Rigden, A. J., Liu, X., Wang, D., Malyshev, S., & Shevliakova, E. (2019a). Urban heat Island: Aerodynamics or imperviousness? *Science Advances*, 5(4), 1–5. <https://doi.org/10.1126/sciadv.aau4299>
- Li, Y., Wang, L., Liu, M., Zhao, G., He, T., & Mao, Q. (2019b). Associated determinants of surface Urban Heat Islands across 1449 Cities in China. *Advances in Meteorology*. <https://doi.org/10.1155/2019/4892714>
- Malik, S., Pal, S. C., Sattar, A., Singh, S. K., Das, B., Chakraborty, R., & Mohammad, P. (2020). Trend of extreme rainfall events using suitable global circulation model to combat the water logging condition in Kolkata Metropolitan Area. *Urban Climate*, 32, 100599. <https://doi.org/10.1016/j.uclim.2020.100599>
- Mann, H. B. (1945). Nonparametric tests against trend. *Econometrica*, 13(3), 245–259. <https://doi.org/10.1016/j.annrmp.2004.07.001>
- Mathew, A., Khandelwal, S., & Kaul, N. (2016). Spatial and temporal variations of Urban Heat Island effect and the effect of percentage impervious surface area and elevation on land surface temperature: Study of Chandigarh City, India. *Sustainable Cities and Society*, 26, 264–277. <https://doi.org/10.1016/j.scs.2016.06.018>
- Mehrotra, Surabhi, Subramanian, Divya, Bardhan, Ronita, & Jana, Arnab. (2021). Effect of surface treatment and built form on thermal profile of open spaces: A case of Mumbai, India. *Urban Climate*, 35(November 2020), 100736. <https://doi.org/10.1016/j.uclim.2020.100736>
- Mirzaei, P. A. (2015). Recent challenges in modeling of Urban Heat Island. *Sustainable Cities and Society*, 19(JUNE), 200–206.
- Mohammad, P. & Goswami, A. (2020). Surface Urban Heat Island variation over Major Indian Cities across Different Climatic Zone. In: *EGU General Assembly Conference Abstracts*, 6444. held online 4–8 May, 2020: 22nd EGU General Assembly. doi: 2020EGUGA..22.6444M.
- Mohammad, P., & Goswami, A. (2019). Temperature and precipitation trend over 139 Major Indian Cities: An assessment over a century. *Modeling Earth Systems and Environment*, 5(4), 1481–1493. <https://doi.org/10.1007/s40808-019-00642-7>
- Mohammad, P., & Goswami, A. (2021a). A Spatio-temporal assessment and prediction of Surface Urban heat island intensity using multiple linear regression techniques over Ahmedabad City, Gujarat. *Journal of the Indian Society of Remote Sensing*, 49(5), 1091–1108. <https://doi.org/10.1007/s12524-020-01299-x>
- Mohammad, P., & Goswami, A. (2021b). Quantifying diurnal and seasonal variation of surface Urban Heat Island intensity and its associated determinants across different climatic zones over Indian Cities. *Giscience and Remote Sensing*, 58(7), 955–981. <https://doi.org/10.1080/15481603.2021.1940739>
- Mohammad, P., & Goswami, A. (2021c). Spatial variation of surface Urban heat island magnitude along the Urban-rural gradient of four rapidly growing Indian Cities. *Geocarto International*. <https://doi.org/10.1080/10106049.2021.1886338>
- Mohammad, P., Goswami, A., & Bonafoni, S. (2019). The Impact of the land cover dynamics on surface Urban heat island variations in semi-arid cities : A case study in Ahmedabad City, India, using multi-sensor/source data. *Sensors*, 19(17), 3701. <https://doi.org/10.3390/s19173701>
- Mohammad, P., Aghlmand, S., Fadaei, A., Gachkar, S., Gachkar, D., & Karimi, A. (2021). Evaluating the role of the Albedo of material and vegetation scenarios along the Urban Street Canyon for improving pedestrian thermal comfort outdoors. *Urban Climate*, 40, 100993. <https://doi.org/10.1016/j.uclim.2021.100993>
- Mohammad, P., Goswami, A., Chauhan, S., & Nayak, S. (2022). Machine learning algorithm based prediction of land use land cover and land surface temperature changes to characterize the surface Urban heat island phenomena over Ahmedabad City, India. *Urban Climate*, 42(2022), 101116. <https://doi.org/10.1016/j.uclim.2022.101116>
- Mukherjee, S., Joshi, P. K., & Garg, R. D. (2017). Analysis of Urban built-up areas and surface Urban Heat Island using downscaled MODIS derived land surface temperature data. *Geocarto International*, 32(8), 900–918. <https://doi.org/10.1080/10106049.2016.1222634>
- Nath, Bibhash, Ni-Meister, Wenge, & Choudhury, Runti. (2021). Impact of Urbanization on land use and land cover change in Guwahati City, India and its implication on declining groundwater level. *Groundwater for Sustainable Development*, 12(September 2020), 100500. <https://doi.org/10.1016/j.gsd.2020.100500>

- Oke, T. R. (1973). City size and the Urban heat island. *Atmospheric Environment Pergamon Pres*, 7, 769–779. [https://doi.org/10.1016/0004-6981\(73\)90140-6](https://doi.org/10.1016/0004-6981(73)90140-6)
- Oke, T. R. (1988). The Urban energy balance. *Progress in Physical Geography*, 12(4), 471–508. <https://doi.org/10.1177/030913338801200401>
- Peng, S., Piao, S., Ciais, P., Friedlingstein, P., Ottle, C., Bréon, F. M., Nan, H., Zhou, L., & Myneni, R. B. (2012). Surface Urban heat island across 419 global big cities. *Environmental Science and Technology*, 46(2), 696–703. <https://doi.org/10.1021/es2030438>
- Peng, Shijia, Feng, Zhili, Liao, Huixuan, Huang, Bei, Peng, Shaolin, & Zhou, Ting. (2019). Spatial-temporal pattern of, and driving forces for, Urban heat island in China. *Ecological Indicators*, 96(Sep-tember 2017), 127–32. <https://doi.org/10.1016/j.ecolind.2018.08.059>
- Pramanik, S., & Punia, M. (2019). Assessment of Green space cooling effects in dense Urban Landscape: A case study of Delhi, India. *Modeling Earth Systems and Environment*, 5(3), 867–884. <https://doi.org/10.1007/s40808-019-00573-3>
- Rahaman, S., Selim Jahangir, Md., Haque, S., Chen, R., & Kumar, P. (2020). Spatio-temporal changes of green spaces and their impact on Urban Environment of Mumbai, India. *Environment, Development and Sustainability*. <https://doi.org/10.1007/s10668-020-00882-z>
- Raj, S., Paul, S. K., Chakraborty, A., & Kuttippurath, J. (2020). Anthropogenic forcing exacerbating the Urban Heat Islands in India. *Journal of Environmental Management*, 257(2019), 110006. <https://doi.org/10.1016/j.jenvman.2019.110006>
- Ramamurthy, P., & Sangobanwo, M. (2016). Inter-annual variability in Urban heat island intensity over 10 major Cities in the United States. *Sustainable Cities and Society*, 26, 65–75. <https://doi.org/10.1016/j.scs.2016.05.012>
- Rizwan, A. M., Dennis, Y. C. L., & Chunho, L. (2008). A review on the generation, determination and mitigation of Urban Heat Island. *Journal of Environmental Sciences*, 20, 120–128.
- Sen, K. P. (1968). Estimates of the regression coefficient based on Kendall's Tau Pranab Kumar Sen. *Journal of the American Statistical Association*, 63(324), 1379–1389. <https://doi.org/10.1080/01621459.1968.10480934>
- Senay, G. B., Budde, M. E., & Verdin, J. P. (2011). Enhancing the simplified surface energy balance (SSEB) approach for estimating landscape ET: Validation with the METRIC model. *Agricultural Water Management*, 98(4), 606–618. <https://doi.org/10.1016/j.agwat.2010.10.014>
- Sharma, S., Anees, M. M., Sharma, M., & Joshi, P. K. (2021). Longitudinal study of changes in ecosystem services in a City of Lakes, Bhopal, India. *Energy, Ecology and Environment*, 6(5), 408–424. <https://doi.org/10.1007/s40974-020-00199-7>
- Shastri, H., Barik, B., Ghosh, S., Venkataraman, C., & Sadavarte, P. (2017). Flip flop of day-night and summer-winter surface Urban heat island intensity in India. *Scientific Reports*, 7(2016), 1–11. <https://doi.org/10.1038/srep40178>
- Singh, P., Kikon, N., & Verma, P. (2017). Impact of land use change and urbanization on Urban heat island in Lucknow City, Central India. A remote sensing based estimate. *Sustainable Cities and Society*, 32, 100–114. <https://doi.org/10.1016/j.scs.2017.02.018>
- Stewart, I. D., & Oke, T. R. (2012). Local climate zones for Urban temperature studies. *Bulletin of the American Meteorological Society*, 93(12), 1879–1900. <https://doi.org/10.1175/BAMS-D-11-00019.1>
- Sultana, S. & Satyanarayana, A. N. V. (2019). Impact of Urbanisation on Urban heat island intensity during summer and winter over Indian Metropolitan Cities. *Environmental Monitoring and Assessment* 191.
- United Nations. (2018). Department of Economic and Social Affairs, Population Division.
- Veena, K., Parammasivam, K. M., & Venkatesh, T. N. (2020). Urban heat island studies: Current status in India and a comparison with the international studies. *Journal of Earth System Science*. <https://doi.org/10.1007/s12040-020-1351-y>
- Voogt, J. A., & Oke, T. R. (2003). Thermal remote sensing of Urban Climates. *Remote Sensing of Environment*, 86(3), 370–384. [https://doi.org/10.1016/S0034-4257\(03\)00079-8](https://doi.org/10.1016/S0034-4257(03)00079-8)
- Wan, Z. (1996). A generalized split-window algorithm for retrieving land-surface temperature from space. *IEEE Transactions on Geoscience and Remote Sensing*, 34(4), 892–905. <https://doi.org/10.1109/36.508406>
- Wan, Z. (2014). New refinements and validation of the collection-6 MODIS land-surface temperature/emissivity product. *Remote Sensing of Environment*, 140, 36–45. <https://doi.org/10.1016/j.rse.2013.08.027>
- Weng, Q. (2009). Thermal infrared remote sensing for Urban climate and environmental studies: methods, applications, and trends. *ISPRS Journal of Photogrammetry and Remote Sensing*, 64(4), 335–344. <https://doi.org/10.1016/j.isprsjprs.2009.03.007>

- World Urbanization Prospects: The 2018 Revision, Online Edition Available online: <https://population.un.org/wup/Publications/Files/WUP2018-Report.pdf> (accessed on 23 Dec 2020)
- Wu, X., Wang, G., Yao, R., Wang, L., Deqing, Yu., & Gui, X. (2019). Investigating surface Urban heat islands in South America based on MODIS data from 2003–2016. *Remote Sensing*, 11(10), 1212. <https://doi.org/10.3390/rs11101212>
- Yadav, N., Chhemendra, S., Peshin, S. K., & Renu, M. (2017). Study of Intra-city Urban heat island intensity and its influence on atmospheric chemistry and energy consumption in Delhi. *Sustainable Cities and Society*, 32(April), 202–11. <https://doi.org/10.1016/j.scs.2017.04.003>
- Yao, R., Wang, L., Wang, S., Wang, L., Wei, J., & Li, J. (2020). A Detailed comparison of MYD11 and MYD21 land surface temperature products in Mainland China. *International Journal of Digital Earth*. <https://doi.org/10.1080/17538947.2019.1711211>
- Yuan, F., & Bauer, M. E. (2007). Comparison of impervious surface area and normalized difference vegetation index as indicators of surface Urban heat island effects in Landsat imagery. *Remote Sensing of Environment*, 106(3), 375–386. <https://doi.org/10.1016/j.rse.2006.09.003>
- Yuanzheng, L. I., Wang, L., Zhang, L., Min, L. I. U., & Zhao, G. (2019). Monitoring intra-annual spatiotemporal changes in Urban heat islands in 1449 cities in China based on remote sensing. *Chinese Geographical Science*, 29(December), 905–16. <https://doi.org/10.1007/s11769-019-1080-9>
- Zhou, D., Zhao, S., Liu, S., Zhang, L., & Zhu, C. (2014). Surface Urban heat island in China's 32 major cities: spatial patterns and drivers. *Remote Sensing of Environment*, 152, 51–61. <https://doi.org/10.1016/j.rse.2014.05.017>
- Zhou, D., Zhao, S., Zhang, L., Sun, Ge., & Liu, Y. (2015). The footprint of Urban heat island effect in China. *Scientific Reports*, 5, 2–12. <https://doi.org/10.1038/srep11160>
- Zhou, D., Xiao, J., Bonafoni, S., Berger, C., Deilami, K., Zhou, Y., Froking, S., Yao, R., Qiao, Z., & Sobrino, J. A. (2019). Satellite remote sensing of surface Urban heat islands: Progress, challenges, and perspectives. *Remote Sensing*, 11(1), 1–36. <https://doi.org/10.3390/rs11010048>

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